

Assignment “Protein structure”

Task: identify the conformation of the AlphaFold predicted model of the human delta-opioid receptor and structural differences between active and inactive conformations using structural alignment.

Context

Opioid receptors are GPCRs (G-protein coupled receptors) involved in analgesia. There are several types of opioid receptors, the major types are mu, delta and kappa opioid receptors. When opioid painkillers (like Vicodin, to which Dr. House was addicted) bind to the opioid receptor, they switch it to active conformation. The change in receptor conformation leads to the dissociation of one of the subunits of the G-protein, which in turn triggers downstream processes that lead to pain relief. Despite being effective analgesics, opioid painkillers have severe side effects such as respiratory depression, constipation, tolerance and dependence. Major efforts are attempted to design safer opioids. One of the approaches is to develop agonists that selectively bind to one type of opioid receptor and don't bind to another.

1. Find the Uniprot accession number for human delta-opioid receptor

Submit the Uniprot accession number of the human delta-opioid receptor.

P41143

2. Find the PDB structures of the protein by Uniprot accession number

How many structures are there for the human DOR in PDB? (1 point)

9 experimental structures + 1 computed structure model

3. Determine the conformation of the receptors

What conformations you identified for each of the three structures? Justify your conclusions. (1 point)

1. 4N6H inactive | because bound to antagonist EJ4 (Naltrindole)
2. 4RWD inactive | because bound to antagonist tetrapeptide H-Dmt-Tic-Phe-Phe-NH₂ (DIPP-NH₂)
3. 6PT2 active | because bound to agonist peptide KGCHM07

4. Explore the AlphaFold model of the human DOR

Based on the global pLDDT metric, what is the overall quality of the model? Comment on all low confidence regions of the structure — hypothesize why AlphaFold was not sure about them. (1 point)

For our structure model, the global pLDDT = 80.4.

From the AlphaFold documentation: 70–90: backbone usually correct; side chains and fine details may be off.

It means that the quality is overall good but could be wrong in details. From pLDDT distribution along sequence it is seen that the model confidence is the lowest at the sequence terminal and loop regions, especially at the terminal regions where disorder is high.

N-terminus is extracellular and C-terminus is a cytoplasmic tail, probably unstructured.

It could be that those regions are intrinsically disordered and need some specific conditions to stabilize in a final conformation, only obtainable in the cellular environment. For example, intracellular regions undergo large rearrangements between inactive and active states and directly contact the G-protein. Without an explicit signaling partner, AlphaFold is unsure which active/inactive geometry to choose.

5. Structural alignment of active and inactive DOR PDB structures and AlphaFold model of DOR

Based on the metrics of the structural alignment, make a conclusion about the conformation of the AlphaFold model of DOR. (1 point)

4N6H (inactive) to AF distance is 19.69 Angstrom.

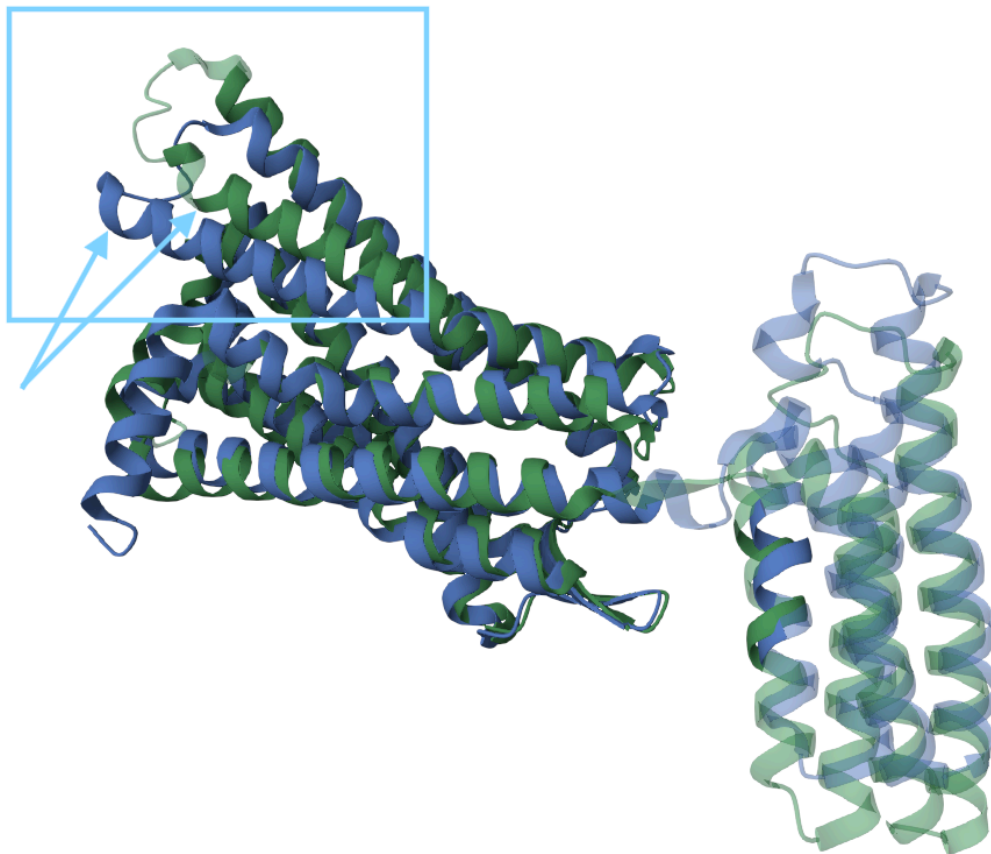
6PT2 (active) to AF distance is 25.68 Angstrom.

Based on these metrics as well as on visual inspection, AlphaFold model conformation looks closer to inactive state.

What region undergoes major rearrangement upon transition from inactive state to active?

- *Submit the picture of the alignment, highlighting this region (in Paint or whatever). (1 point)*
- *Refer to the literature on the difference between active and inactive states of GPCRs to check that you identified the region correctly. Provide a quote and a link to the source. (1 point)*

Picture of active vs inactive states alignment:



Blue arrows highlight TM6 helices at the cytosolic side from active and inactive states. It seems to change the most between those two states, as well as the adjacent loop and TM5. But overall all helices seem to change, with the most change in TM6.

Quote: “We find that outward movement and rotation of TM6 on the cytosolic side—opening for G-protein coupling—is a universal feature of activation throughout the GPCR superfamily”.

Link: [GPCR activation mechanisms across classes and macro/microscales - PMC](#)